



VISUAL & INTUITIVE EDITION

The Cognitive Skills Library

One skill per page with visual diagrams, everyday human analogues, and realistic software missions. Turning prompt chaos into rock-solid autonomous intelligence!

18**VISUAL RUNBOOKS**

1 dedicated full-page guide per skill

100%**REAL EXAMPLES**

Concrete prompt & execution traces

Zero**BORING JARGON**

Intuitive analogies & graphical flows

**WHY THIS LIBRARY EXISTS**

LLMs don't fail because they aren't smart. They fail because raw prompts are like shouting instructions across a noisy kitchen!

A **Skill** is a recipe card with strict ingredients, numbered steps, and a test at the end to make sure nothing burned. When you package cognition as skills, models become dependable partners that plan ahead, double-check their math, and never hallucinate!



Master Table of Contents

Every skill is presented on its own full page with dedicated visual diagrams and walkthroughs:

1. Executive Function 1.1 goal-decomposition 1.2 dynamic-replanning 1.3 deliberate-depth-router 1.4 inhibitory-control-gate 4 Skills	2. Invariant Reasoning 2.1 first-principles-reduction 2.2 counterfactual-stress-testing 2.3 abductive-diagnostician 2.4 analogical-domain-transfer 4 Skills
3. Metacognition 3.1 adversarial-self-critique 3.2 uncertainty-calibration 3.3 cognitive-backtracking 3 Skills	4. Memory Systems 4.1 working-memory-compactor 4.2 episodic-distiller 4.3 semantic-graph-linker 3 Skills
5. Theory of Mind 5.1 theory-of-mind-profiler 5.2 consensus-synthesizer 2 Skills	6. Divergent Thinking 6.1 lateral-explorer 6.2 boundary-fuzzer 2 Skills

EXECUTIVE FUNCTION

PAGE 3

#1.1 goal-decomposition

🌟 Turn messy wishes into clean, step-by-step game plans!

🧠 THE EVERYDAY HUMAN SUPERPOWER

How our brain takes 'cook Thanksgiving dinner' and turns it into shopping, prepping turkey, baking pies, and timing the oven.

When AI fires this skill: User gives a big, vague, or multi-step goal ('Build me an auth system', 'Migrate database to Postgres').

📊 PROCEDURAL VISUAL FLOW

Stripe Webhooks



DB Schema



Checkout UI



E2E Test

🎯 Concrete Mission: Real-World Mission: Launch a Stripe Subscription Flow

"Hey Claude, let's build recurring billing for our SaaS app by Friday."

- Step 1: Setup Stripe webhook listener endpoint & test idempotency.
- Step 2: Create Postgres 'subscriptions' table with foreign keys.
- Step 3: Wire frontend checkout session redirect with error fallback.
- Step 4: Run end-to-end sandbox purchase test to verify green status.



The Golden Rule: Never start coding step 3 until step 1 & 2 have hard evidence they work!

EXECUTIVE FUNCTION

PAGE 4

#1.2 dynamic-replanning

🗺️ GPS rerouting for autonomous agents when roads get blocked!

🧠 THE EVERYDAY HUMAN SUPERPOWER

When Google Maps sees highway construction ahead and instantly finds an alternate route without making you drive back home.

When AI fires this skill: An API throws a 403 error, a unit test fails unexpectedly, or a needed file is missing.

📊 PROCEDURAL VISUAL FLOW

Task A (Done)



Task B (403 Error!)



⚡ Auto-Reroute to GraphQL



Task C (Success)

🎯 Concrete Mission: Real-World Mission: API Rate Limit Crashes Mid-Scrape

"Error: GitHub API 403 Secondary Rate Limit Exceeded on node #4."

- Detect: Node #4 failed due to transient rate limits.
- Prune: Don't kill the entire 20-node data extraction pipeline!
- Reroute: Switch node #4 to GraphQL batching + cached mirrors.
- Resume: Re-splice new branch; pipeline finishes without restarting.



The Golden Rule: Don't restart the whole task from scratch when only one branch stumbled!

EXECUTIVE FUNCTION

PAGE 5

#1.3 deliberate-depth-router

🚀 Fast reflexes for easy questions, deep thinking for high-stakes decisions!

🧠 THE EVERYDAY HUMAN SUPERPOWER

Catching a falling cup is instant System 1 reflex; planning retirement investments is deep System 2 deliberation.

When AI fires this skill: Incoming task arrives. Evaluates risk: Is it read-only, or could it delete production data?

📊 PROCEDURAL VISUAL FLOW

User Intent



⚡ Low Risk: Fast Reflex



🧠 High Risk: 5-Step Deep Search

🎯 Concrete Mission: Real-World Mission: 'What's the weather?' vs. 'Drop production DB'

Prompt A: "Summarize this README file." vs. Prompt B: "Drop customer table and reseed."

- Prompt A: Fast Path! 0.2s response, minimal token spend, simple summary.
- Prompt B: Red Alert System 2! Pause, verify backup snapshots, require confirmation.



The Golden Rule: Save compute on trivial lookups so you have maximum brainpower for critical code!

EXECUTIVE FUNCTION

PAGE 6

#1.4 inhibitory-control-gate

👉 The ultimate impulse filter: Stop hallucinations before they speak!

🧠 THE EVERYDAY HUMAN SUPERPOWER

Biting your tongue before saying something you know you will immediately regret.

When AI fires this skill: Model generates code or text containing fake URLs, leaked secrets, or unauthorized shell commands.

📊 PROCEDURAL VISUAL FLOW

Agent Impulse



🛡️ Security & Truth Gate



Verified Safe Output

🎯 Concrete Mission: Real-World Mission: Model Tries to Call a Fake API Function

Agent attempts: ``await stripe.instant_magical_wire_transfer()``

- Filter: Gate scans schema library. Function does not exist!
- Suppression: Block action instantly before sending network packet.
- Correction: Query official Stripe SDK docs; replace with ``stripe.transfers.create()``.



The Golden Rule: A great AI knows not just what to say, but what to suppress!

INVARIANT REASONING

PAGE 7

#2.1 first-principles-reduction

✨ Strip away all the hype and jargon until only pure physics & math remain!

THE EVERYDAY HUMAN SUPERPOWER

Elon Musk asking: 'What does a battery truly cost if we only buy cobalt, lithium, and nickel on the London metal exchange?'

When AI fires this skill: Told 'That's impossible' or staring at an over-engineered legacy system with 50 bloated microservices.

PROCEDURAL VISUAL FLOW

Bloated Assumptions



Core Axioms (Math & Physics)



Simple, High-Speed
Architecture

Concrete Mission: Real-World Mission: Rebuilding a \$50k/mo Cloud Queue with SQLite

"Our Kafka cluster is costing a fortune just to process 500 events a minute."

- Strip hype: We don't need distributed partitioning for 8 events per second.
- Axiom: An append-only local SQLite file writes 50,000 tx/sec with zero latency.
- Result: Replaced 12 cloud nodes with 1 file. Latency dropped 90%, cost dropped to \$0.



The Golden Rule: When stuck, boil the problem down to elementary truths and reason up from there.

#2.2 counterfactual-stress-testing

👉 Playing devil's advocate: 'What if everything that can go wrong, does?'

🧠 THE EVERYDAY HUMAN SUPERPOWER

Flight simulators testing pilots by suddenly killing engine #1 and freezing the altimeter in mid-air.

When AI fires this skill: About to approve a new system architecture, security protocol, or deployment script.

📊 PROCEDURAL VISUAL FLOW

Base Design



⚡ What if DB times out?



🛡️ Circuit Breaker Defenses Added

🎯 Concrete Mission: Real-World Mission: Pre-Mortem on a Black Friday Checkout API

"Architecture proposal: All checkout events write directly to primary Postgres."

- Scenario 1: What if traffic spikes 50x in 3 seconds? -> Connection pool exhausted!
- Scenario 2: What if Redis cache dies? -> DB throttles and 504 errors skyrocket.
- Defense Injected: Circuit breaker + async Redis buffer + graceful queue degradation.



The Golden Rule: Break your own system on paper today so customers don't break it on Black Friday.

INVARIANT REASONING

PAGE 9

#2.3 abductive-root-cause-diagnostician

🕵️ Sherlock Holmes for code: Deduce the real culprit behind mysterious bugs!

🧠 THE EVERYDAY HUMAN SUPERPOWER

A doctor diagnosing why a patient has fever by systematically ruling out flu, allergies, and infections.

When AI fires this skill: Spotted a cryptic stack trace, intermittent race condition, or memory leak in server logs.

📊 PROCEDURAL VISUAL FLOW

Mystery Crash

Candidate
HypothesesPopperian
FalsificationConfirmed Root
Cause

🎯 Concrete Mission: Real-World Mission: The Server Crashes Every Tuesday at 3:00 AM

"Logs show Out-Of-Memory kill every Tuesday 03:02 UTC with no user traffic."

- Hypothesis A: Memory leak in worker pool? (Falsified: Memory flat on other days).
- Hypothesis B: Cron job dumping unindexed backup into RAM? (Verified: 10GB dump at 03:00!).
- Cure: Stream backup via disk pipe instead of buffering in Node.js buffer.



The Golden Rule: Don't guess or randomly tweak code. Formulate hypotheses and actively try to disprove them.

#2.4 analogical-domain-transfer

👉 Steal brilliant solutions from biology, music, and physics to solve code!

🧠 THE EVERYDAY HUMAN SUPERPOWER

How Velcro was invented by studying how burdock burrs stick to dog fur on hiking trails.

When AI fires this skill: Tackling an unfamiliar software problem with no obvious library or tutorial available.

📊 PROCEDURAL VISUAL FLOW

Unsolved Problem



💡 Analogy from Nature / Arts



Novel Software Pattern

🎯 Concrete Mission: Real-World Mission: Managing Multi-Agent Chat Traffic via Symphony Orchestration

"How do we coordinate 10 AI agents without them talking over each other in chat?"

- Analogy: An orchestral symphony conductor.
- Concept: Conductor holds the baton (Token lock). Woodwinds play only when cued.
- Result: Built a Conductor-Agent pattern that grants turn-by-turn speech tokens cleanly.



The Golden Rule: The best code architectures are often disguised copies of natural systems that already work.

METACOGNITION

PAGE 11

#3.1 adversarial-self-critique

👉 Send in the internal Red Team to punch holes before the customer sees it!

🧠 THE EVERYDAY HUMAN SUPERPOWER

An author having their harshest editor read the book draft before sending to the publisher.

When AI fires this skill: Immediately after drafting code, SQL queries, or architectural specifications.

📊 PROCEDURAL VISUAL FLOW

First Draft

 Adversarial Critique Pass

Polished & Hardened Artifact

🎯 Concrete Mission: Real-World Mission: Auditing a New User Sign-Up API Endpoint

Drafted code: `INSERT INTO users (email, password) VALUES (\$1, \$2)`

- Red-Team Critique 1: Password is in plaintext! Must bcrypt with 12 salt rounds.
- Red-Team Critique 2: No check for lowercase email normalization! Duplicates possible.
- Red-Team Critique 3: Missing rate limit! Vulnerable to bot signup spam.
- Correction Applied: All 3 patched before output was ever emitted.



The Golden Rule: Catch your own mistakes in private so you never have to apologize for them in public.

METACOGNITION

PAGE 12

#3.2 epistemic-uncertainty-calibration

🧙 Knowing what you don't know: Trade guessing for accurate research!

🧠 THE EVERYDAY HUMAN SUPERPOWER

A world-class pilot checking the radar manual instead of guessing how a button works in a storm.

When AI fires this skill: Model generates facts about recent libraries, API parameters, or pricing numbers.

📊 PROCEDURAL VISUAL FLOW

Question



Epistemic Confidence Meter

High: Answer | Low: Query
Docs First

🔗 Concrete Mission: Real-World Mission: Configuring Next.js 15 Server Actions

User asks: "What is the exact signature for Server Actions revalidation in Next.js 15?"

- Entropy Check: Confidence is 65% (Next.js 15 introduced breaking changes).
- Rule: Confidence < 80% triggers mandatory tool lookup!
- Action: View Next.js official migration doc on disk instead of hallucinating.
- Result: Delivers 100% verified signature with zero syntax errors.



The Golden Rule: Confidence isn't bravado. Real intelligence is measuring your own doubt and looking up the truth.

METACOGNITION

PAGE 13

#3.3 cognitive-state-backtracking

🔑 Git Revert for your brain: Rewind cleanly when a path hits a dead end!

🧠 THE EVERYDAY HUMAN SUPERPOWER

Exploring a maze: When you hit a brick wall, you walk back to the last fork instead of bashing your head against the wall.

When AI fires this skill: An implementation approach fails after 3 attempts or breaks existing test suites.

📊 PROCEDURAL VISUAL FLOW

Start



Path A (Dead End!)

 Rewind to ForkPath B (Clean
Victory)

🎯 Concrete Mission: Real-World Mission: Refactoring a Monolith Module to Rust

"Compiler error: Lifetimes cannot be satisfied in async stream across thread boundaries."

- Recognize: This specific async lifetime design is an architectural dead end.
- Avoid: Don't spend 4 hours adding messy `Arc` hacks everywhere.
- Rewind: Rollback git branch to checkpoint #3; switch to actor-based message passing.
- Success: Clean compilation in 15 minutes.



The Golden Rule: Never fall in love with a bad idea just because you spent time on it. Roll back and take the open door.

#4.1 working-memory-compactor

🌟 KonMari your context window: Spark joy, ditch clutter, keep the essentials!

🧠 THE EVERYDAY HUMAN SUPERPOWER

Writing down key phone numbers on an index card so your desk isn't buried in 500 sticky notes.

When AI fires this skill: Conversation tokens cross 65% limit or multi-file debugging session gets noisy.

📊 PROCEDURAL VISUAL FLOW

Messy 80k Context



🔧 Compaction Filter



Tidy 1.5k Active Scratchpad

🎯 Concrete Mission: Real-World Mission: Compacting a 70,000-Token Debugging Marathon

History has 40 pages of repetitive npm error logs, solved tickets, and chatter.

- Evict: Wipe away 35 pages of raw error logs and solved intermediate bugs.
- Retain: Active DB schema, currently failing unit test, and active API keys.
- Yield: Context drops from 70,000 tokens to 1,200 tokens. Speed triples!



The Golden Rule: An agent with a clean context thinks 10x faster and never forgets the core goal.

MEMORY SYSTEMS

PAGE 15

#4.2 episodic-retrospective-distiller

🌟 Turn hard-won lessons into permanent recipes so you never fail twice!

🧠 THE EVERYDAY HUMAN SUPERPOWER

Grandma perfecting her secret cookie recipe card after burning the first batch.

When AI fires this skill: Successfully solved a difficult bug after multiple attempts or designed a novel workflow.

📊 PROCEDURAL VISUAL FLOW

2 Hours of Trial & Error



💡 Extract Working Pattern



Permanent SKILL.md Recipe

🎯 Concrete Mission: Real-World Mission: Auto-Synthesizing a New Docker Deployment Skill

"We spent 2 hours figuring out why Apple Silicon M3 failed to build x86 Docker images."

- Lesson Learned: `docker buildx` requires `--platform linux/amd64` flag and Rosetta emulation.
- Distillation: Auto-generates `SKILL.md` under `workspace/skills/m3-docker-build/`.
- Result: Next time user says 'Build docker image', agent executes perfect flag automatically.



The Golden Rule: Every painful mistake is a golden opportunity to write a permanent procedural skill.

#4.3 semantic-graph-linker

🔗 **Connect the dots: Build a 3D web of facts instead of loose keyword piles!**

🧠 THE EVERYDAY HUMAN SUPERPOWER

Detective boards with red string connecting suspects, dates, locations, and motives.

When AI fires this skill: Reading large codebases, multi-repo projects, or extensive architectural documentation.

📊 PROCEDURAL VISUAL FLOW

Auth Service



Billing Service



Redis Cache



Kafka Queue

🎯 Concrete Mission: Real-World Mission: Mapping Enterprise Microservices Relationships

"Doc says Auth Service talks to Redis, Billing reads Auth, and Notification consumes Kafka."

- Extract Triples: `(BillingService) -[VALIDATES\_VIA]-> (AuthService)`
- Extract Triples: `(AuthService) -[CACHES\_SESSIONS\_IN]-> (RedisCluster)`
- Query: 'What breaks if Redis restarts?' Graph instantly traces: Auth & Billing!



The Golden Rule: Vectors find similar text; Knowledge Graphs find exact dependencies and causal ripple effects.

#5.1 theory-of-mind-profiler

🌟 **Read between the lines: Understand what the user really wants, not just what they typed!**

🧠 THE EVERYDAY HUMAN SUPERPOWER

A senior barista who knows you need an espresso double-shot without you having to explain your morning.

When AI fires this skill: User gives short, ambiguous, or highly technical prompts without full background context.

📊 PROCEDURAL VISUAL FLOW

Terse User Prompt



🧠 Latent Intent & Persona Calibration



Perfect-Fit Execution

🎯 Concrete Mission: Real-World Mission: User asks 'Make it fast'

"Hey, make this page fast."

- Profile: User is a Staff Frontend Engineer shipping to mobile 3G devices.
- Latent Intent: They care about Largest Contentful Paint (LCP) and bundle size, not DB indexing.
- Execution: Optimize image formats (WebP), defer heavy JS scripts, tree-shake CSS.



The Golden Rule: Amateurs answer literal words; elite AI answers the unstated human intent.

#5.2 dialectical-consensus-synthesizer

🧠 Turn heated arguments into brilliant hybrid breakthroughs!

🧠 THE EVERYDAY HUMAN SUPERPOWER

Two master chefs arguing about sweet vs. savory and inventing salted caramel.
When AI fires this skill: Two AI agents (or team members) give polar opposite technical proposals.

📊 PROCEDURAL VISUAL FLOW

Thesis (Fast)



Antithesis (Secure)



🧠 Higher-Order Synthesis
(Both!)

🎯 Concrete Mission: Real-World Mission: Speed vs. Security Conflict

Agent A: "Cache everything in browser localStorage!" vs. Agent B: "Never store tokens on client!"

- Thesis (Speed): Zero network latency for cached profiles.
- Antithesis (Security): XSS can steal tokens from localStorage.
- Synthesis: HTTP-only Secure Cookies for auth tokens + localStorage for public UI assets!

🔑 **The Golden Rule:** Don't compromise down to a mediocre middle. Synthesize up to an elegant architecture.

DIVERGENT THINKING

PAGE 19

#6.1 lateral-divergent-explorer

🌟 Think outside the box: Break free from boring, predictable answers!

🧠 THE EVERYDAY HUMAN SUPERPOWER

Jazz musicians improvising unexpected chord progressions to create magic.

When AI fires this skill: Brainstorming new features or when optimization reaches diminishing returns.

📊 PROCEDURAL VISUAL FLOW

Boring Standard Way



🚀 4 Wild Divergent Concepts



Game-Changing
Breakthrough

🎯 Concrete Mission: Real-World Mission: Building a Customer Loyalty Feature

"How do we make our shopping app more engaging than standard discount coupons?"

- Boring Generic: 10% off coupons (Zero user excitement).
- Lateral Leap 1: Mystery treasure chest unlocked by sharing recipes.
- Lateral Leap 2: Team shopping cart where friends unlock bulk discounts together.
- Winner: Team Cart boosted viral invites by 340%!



The Golden Rule: To find a revolutionary idea, you have to be brave enough to sample far outside the comfort zone.

DIVERGENT THINKING

PAGE 20

#6.2 boundary-constraint-fuzzer

🌟 The crazy stress-test: Throw kitchen sinks and alien emojis at your code!

🧠 THE EVERYDAY HUMAN SUPERPOWER

Crash-testing a car into concrete walls at 90mph before putting a family inside.

When AI fires this skill: Writing API endpoint validators, form parsers, or serialization protocols.

📊 PROCEDURAL VISUAL FLOW

User Input Field



💣 Extreme Fuzzing Payloads



🛡️ Bulletproof Production Code

🎯 Concrete Mission: Real-World Mission: Testing a Bank Transfer Amount Field

User inputs amount to transfer: `amount: number`

- Fuzz Test 1: `-100.00` (Did it allow negative transfer stealing money?) -> Blocked!
- Fuzz Test 2: `0.00000000001` (Did float precision truncate to zero?) -> Handled via BigInt!
- Fuzz Test 3: `NaN`, `Infinity`, `null`, `DROP TABLE` -> All rejected gracefully!



The Golden Rule: If you don't fuzz your software with evil inputs, your users will do it for you on day one.